

1. (Fall 2024, #7) Given the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2) \cdot 2^n} (x-5)^n$$

- (a) find its radius of convergence,
- (b) find its interval of convergence (that is, the set of points where it is convergent),
- (c) and find an upper bound for the remainder $|R_5|$ at the only point where the series converges conditionally.

2. (Spring 2024, #8) Consider the function m defined by the power series

$$m(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n}}{2n+1} = 1 - \frac{x^4}{3} + \frac{x^8}{5} - \frac{x^{12}}{7} + \frac{x^{16}}{9} - \dots$$

- (a) Find the radius of convergence of m .

- (b) Find the interval of convergence of m .

- (c) Find $m^{(24)}(0)$.

- (d) Find the value of

$$\int_0^{1/2} m(x) dx$$

with an error of less than $0.001 = 1/1000$. Leave your answer as a finite sum of fractions.

- (e) What is the function $m(x)$? Find an expression for $m(x)$ using familiar functions.

3. (Spring 2024, #10)

(a) A power series $\sum_{n=0}^{\infty} a_n(x-7)^n$ converges when $x = 3$ and diverges when $x = 12$. Does the series converge when $x = 8$? Circle the best option and explain briefly.

- i. It converges absolutely.
- ii. It converges though it may only converge conditionally.
- iii. It diverges.
- iv. There is not enough information to determine this.

(b) Consider the power series $\sum_{n=0}^{\infty} a_n(x-9)^n$. Suppose

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{7^n} = 4.$$

What is the radius of convergence of the power series, R ? Choose one option and explain briefly.

- i. $1/7$
- ii. $1/4$
- iii. 4
- iv. 7
- v. 9
- vi. There is not enough information to determine this.

4. (Fall 2023, #7)

(a) Find the interval of convergence of the series

$$\sum_{n=2}^{\infty} \frac{(-2x)^n}{n^{3/2} \cdot \ln(n+1)}$$

(b) Assume the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ is R , where $a_n \geq 0$. Does the series $\sum_{n=1}^{\infty} a_n \sqrt{n} x^{n-1}$ converge for $0 \leq x < R$? Justify your answer.

5. (Fall 2023, #8)

(a) Find a power series representation of $f(x) = (1+x)^3 \cdot \ln(2+x)$ and its radius of convergence.

(b) Find $f^{(20)}(-1)$.